

Spatial and spatio-temporal methods for mapping malaria risk: a systematic review

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ABSTRACT

Background Approaches in malaria risk mapping continue to advance in scope with the advent of geostatistical techniques spanning both the spatial and temporal domains. A substantive review of the merits of the methods and covariates used to map malaria risk has not been undertaken. Therefore, this review aimed to systematically retrieve, summarise methods and examine covariates that have been used for mapping malaria risk in sub-Saharan Africa (SSA).

Methods A systematic search of malaria risk mapping studies was conducted using PubMed, EBSCOhost, Web of Science and Scopus databases. The search was restricted to refereed studies published in English from January 1968 to April 2020. To ensure completeness, a manual search through the reference lists of selected studies was also undertaken. Two independent reviewers completed each of the review phases namely: identification of relevant studies based on the Preferred Reporting Items for Systematic Reviews and Meta-Analyses guidelines, data extraction and methodological quality assessment using a validated scoring criterion.

Results One hundred and seven studies met the inclusion criteria. The median quality score across studies was 12/16 (range: 7–16). Approximately half (44%) of the studies employed variable selection techniques prior to mapping with rainfall and temperature selected in over 50% of the studies. Malaria incidence (47%) and prevalence (35%) were the most commonly mapped outcomes, with Bayesian geostatistical models often (31%) the preferred approach to risk mapping. Additionally, 29% of the studies employed various spatial clustering methods to explore the geographical variation of malaria patterns, with Kulldorf scan statistic being the most common. Model validation was specified in 53 (50%) studies, with partitioning data into training and validation sets being the common approach.

Conclusions Our review highlights the methodological diversity prominent in malaria risk mapping across SSA. To ensure reproducibility and quality science, best practices and transparent approaches should be adopted when selecting the statistical framework and covariates for malaria risk mapping. Findings underscore the need to periodically assess methods and covariates used in malaria risk mapping; to accommodate changes in data availability, data quality and innovation in statistical methodology.

Summary box

What is already known?

- The disproportionate decline of malaria risk over-time and between/within countries in sub-Saharan Africa attributed to biological, environmental, social and demographic factors has triggered a renewed interest in its fine-scale epidemiology.
- Enhanced computational ability and availability of data of high quality and volume has enabled the quantification malaria risk burden in space and time leading to the proliferation of methods within a formal statistical framework.
- The complexity of spatio-temporal models has increased, making inferential and predictive processes difficult to undertake.

What are the new findings?

- The production of more granular estimates of malaria risk hinges on accessibility to and collection of timely data at finer resolutions.
- Variable selection should be objectively developed to contribute to the maximum predictive accuracy of the spatio-temporal model.

What do the new findings imply?

- Spatio-temporal approaches need to robustly quantify the sub-national burden of malaria risk, as an epidemiological prerequisite to intervention strategies.
- Investments in primary data collection at subnational scales, development and continuous application of robust modelling tools and approaches will be important for orienting malaria control and elimination efforts in the next decade.
- As the malaria landscape diversifies, new tools will be required to not only highlight changes locally, but also to provide evidence-based insights into factors driving the change.

INTRODUCTION

Global efforts to control and eliminate malaria are intrinsically linked to the Sustainable Development Goals.¹ Specifically, the Global Technical Strategy (GTS) for Malaria (2016–2030) reiterates the need to reduce both malaria case incidence and mortality

rates by up to 90%² in high burden countries, mostly concentrated in sub-Saharan Africa (SSA), and eliminating malaria in at least 35 countries and preventing resurgence in malaria-free countries.² However, in 2018, SSA had an estimated 213 million clinical episodes of malaria, caused mainly by *Plasmodium falciparum* parasite³. To address this high burden, the GTS emphasises on the need to target interventions according to subnational disease risk stratification.²

The importance of malaria risk mapping in Africa can be traced back to the mid-1950s when malaria epidemiology formed a critical prelude to the design of interventions aimed at eliminating malaria.⁴ A resurgence in malaria cartography emerged in the 1990s,^{4–6} coinciding with an era of intensive control and elimination activities. Over the last 20 years, national and subnational malaria risk maps have been developed in many endemic countries in SSA.^{4,7,8} This has led to a proliferation of methods and an increase in data quality and quantity—prompted by the demands for robust and reliable characterisation of malaria risk in space and time.

The science of malaria cartography has evolved from hand-drawn risk maps to contemporary digital maps due to the demand for computational solutions and methodologies. These are needed to produce accurate estimates at a high spatial and temporal resolution to facilitate monitoring elimination progress within and between countries in SSA.⁸ Modern mapping embracing novel statistical techniques at high spatial and temporal resolution are increasingly being used to inform public health policy.^{9,10} Most recently, this has been aided by the availability of curated spatial databases, geographical information systems, enhanced computational capabilities and the advancement in spatial statistics. Standardised nationally representative survey initiatives, such as the geolocated Malaria Indicator Survey and the Demographic and Health Survey platforms, have availed geocoded malaria data with relevant covariates.^{11,12} This has enabled the characterisation of malaria risk at a high spatial resolution over which health policy is made.

Previous reviews have been conducted to; identify environmental risk factors of malaria transmission,^{11,13} summarise methodological and computational power advancement.¹² Despite the increase in the number of malaria risk mapping studies, there are no recent and comprehensive reviews of the changes in methodological frameworks and covariates used. Consequently, we aimed to identify and review malaria risk mapping studies, to assess analytical methods and covariates used in the last five decades.

METHODS

The protocol guiding this review has been previously published.¹⁴ Our results are reported according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses guidelines.^{15,16} A notable deviation from our protocol was limiting the review scope to SSA where

the burden of malaria is highest and countries have broadly similar malaria vector and parasite ecologies and health system contexts, compared with low-income and middle-income countries.^{3,17} A rigorous three-phase process was undertaken to transparently identify and summarise spatio-temporal studies based on their methodical framework and covariates employed used in malaria risk mapping.

Phase 1: Identification of relevant studies/keyword search.

Search terms and databases

All studies published between 1 January 1968 and 30 April 2020 were systematically searched through four electronic databases (PubMed, Web of Science, EBSCOhost and Scopus) using search terms defined in online supplemental table 1. To improve the search strategy, thematically mined keywords were funnelled using Boolean operators and truncations before being employed across the selected electronic reference databases. The starting year (1968) corresponded to the year when the first global audit of malaria endemicity was undertaken.^{5,12} Relevant studies were imported into Endnote, version X9 (Clarivate Analytics, Philadelphia, Pennsylvania, USA) (online supplemental table 1).

Phase 2: Study selection

Studies were screened independently by two authors JNO and CK for possible inclusion based on information provided in the title and abstract. Relevant studies based on the research questions were subsequently appraised on their eligibility for full-text review. The full-text review entailed the application of a more stringent inclusion/exclusion criteria for selecting studies to be included for data extraction. Additional papers were identified by examining the reference lists of retrieved studies and by contacting the authors where necessary. Emerging discrepancies were resolved by consensus and by an independent arbitrator (BS). A comprehensive and pilot tested form was used for data extraction.

Inclusion and exclusion criteria

Peer-reviewed studies that employed spatial, temporal and spatio-temporal modelling techniques in malaria risk mapping in SSA were considered. A spatial model was defined as one that explicitly included a geographical index, while a temporal model included a time index. Studies using at least one visualisation or modelling technique (with or without covariates) for assessing the burden of malaria were included. Commentaries, expert reviews and/or reports that did not include original research were read, and only relevant studies cited included.

Phase 3: Data extraction

A standardised extraction form was used to independently extract the data by two reviewers (JNO and CK). The tool was first piloted and refined accordingly. Discordance between the reviewers with respect to the information

extracted was resolved by consensus and by consulting with an independent arbitrator (BS). For each selected study, the following information was extracted (online supplemental table 2) namely:

- Bibliographic information (Author, year, study setting and period, primary unit of analysis, spatial and temporal resolution).
- Study objective(s)
- Data aspects (data sources, malaria data, covariate type)
- Analytical method (modelling approach(es), assumptions, cluster detection techniques, statistical tests, diagnostic/validation checks).
- Results and discussions (key findings, modelling gaps, recommendation(s)).

Quality assessment

A previously used 8-point scoring criteria¹⁸ was adapted and modified to assess the quality of the individual studies based on their aims and objectives, input data, model validity, results and conclusions (online supplemental table 3). Screening questions/criterion were used to guide the scoring process, with the score ranging from 0 (poor) to 2 (good) on each criterion. The overall quality level assigned to individual studies were summarised into four broad categories; very high (>13), high (11–13), medium (8–10) and low (<8) (online supplemental table 3).

RESULTS

Literature search, data synthesis and quality assessment

A total of 7189 studies were retrieved from the various databases with 170 studies fully screened after the title and abstract review. Ultimately 107 studies were included for

review and underwent quality assessment and synthesis (figure 1).

The distribution of studies by geographical scale and scope varied across SSA. Five (5%) studies were continental in scale, 48 (45%) studies were national and 52 (49%) studies were subnational. Kenya (10 studies) and Tanzania (9 studies) had the highest number of publications included in the review (figure 2).

The longest study period spanned 115 years,¹⁹ while the shortest study period was 3 months.²⁰ Fifty-eight (54%) studies had an overall study period ranging between 3 months and 5 years, while 21 (20%) studies had their study period ranging between 6 and 10 years and 28 (26%) studies spanned more than 10 years. Overall, the number of publications increased over the review period (figure 3).

The median score was 12 out of 16, with 16 representing the highest possible quality. The overall quality score of the reviewed studies ranged from 7 to 16. Two studies were of low quality, 22 studies were of medium quality, 42 studies were of high quality and 41 studies were of very high quality (online supplemental table 3).

Data sources, covariate selection and preprocessing

From the review, global, continental, national and subnational databases/repositories provided a rich source of both malaria data and covariates used for modelling. These sources comprised of geographically referenced surveys used by 34 (32%) studies, 20 (19%) studies used population databases and 10 (9%) studies used government records. Routinely collected data from the Health and Demographic Surveillance System were used in 16 (15%) studies. Sources of climatic and environmental covariates consisted of ground station observations used by 17 (16%) studies and remotely sensed satellite surrogates of climate, urbanisation and topography were employed by 49 (46%) studies (table 1).

In this review, variable selection techniques were explicitly specified by 47 (44%) studies. These techniques varied substantially; with the frequentist approach used in 14 (13%) studies to assess the (uni and multi) variate association between malaria outcomes and its covariates being the most common. Significant covariates were included if their nominal p value was less than 0.001,^{21–24} 0.05,^{25–27} 0.1,²⁸ 0.15,^{29–30} 0.2,^{31–32} and 0.25.³³ The generalised linear models used by nine (8%) studies identified the best covariate subset based on Wald's p value^{20–32–34–36} and the variance inflation factor.³⁷ Additionally, six (6%) studies used the total-sets analysis based on Bayesian information criterion (BIC) statistic to identify the optimal variable combination. Principal component analysis was employed by eight (8%) studies to reduce dimensions and avoid collinearities in environmental factors,^{38–39} meteorological factors^{38–40} and household demographics.^{32–33–41–43} The Bayesian stochastic search was used by three (3%) studies to identify covariates with the highest inclusion probability. Other techniques employed included the least absolute shrinkage and selection operator (LASSO)

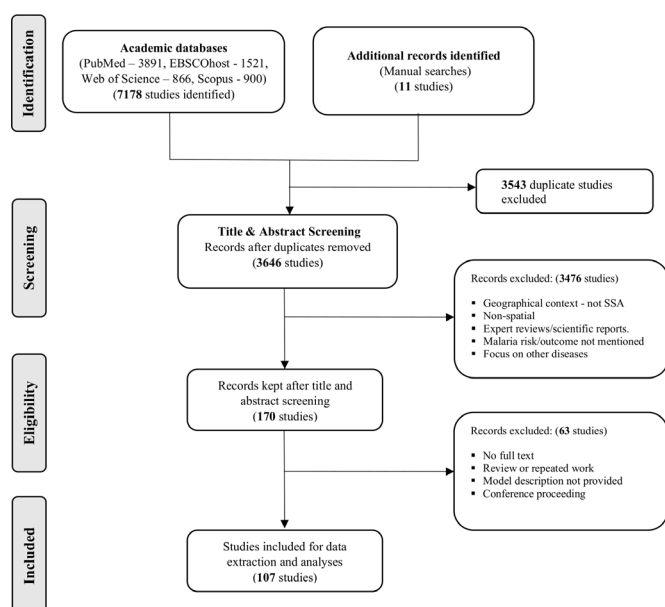


Figure 1 Study flow from literature search to data extraction and analyses.

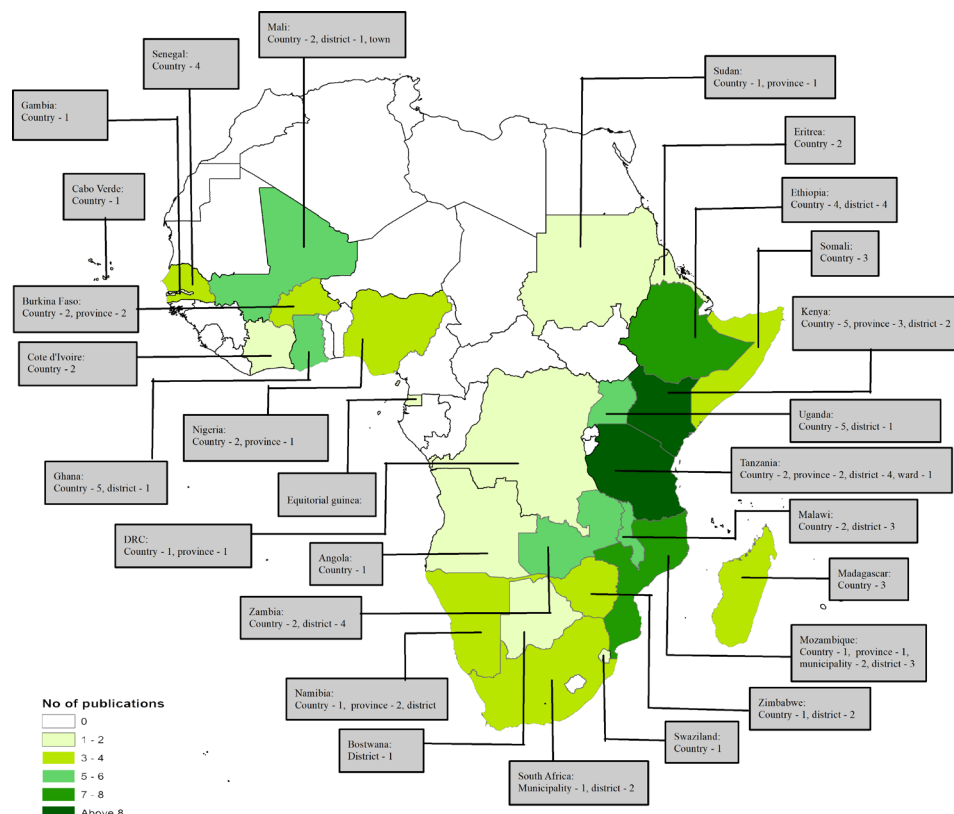


Figure 2 Geographical scale and scope of studies. Geographical scale (municipality, district, province/state, country) of studies is given in grey boxes. The studies covered 27 countries in sub-Saharan Africa with East Africa being the most represented subregion.

penalty, the Spike and slab and the Bayesian model averaging cumulatively used by five (5%) studies. Two studies (2%) reviewed covariates used in past studies to identify and adopt the best suite of covariates to be included in their model (table 2).

Data preprocessing procedures were employed in 37 (35%) studies. The verification of geographical coordinates by either paper maps⁴⁴ or global digital

maps^{8 31 34 36 42 44-58} used in 20 (21%) studies was the most common procedure. Algorithms based on the catalytic conversion models^{8 36 46 48-50 55 59-61} were used in 10 (9%) studies to generate age-adjusted malaria prevalence predictions, that is, age range of 2–10 years. Continuous variables were standardised in seven (7%) studies; by log transformation,⁶² centring on the mean^{47 63-66} and zero.⁴⁷ Four (4%) studies excluded study regions with inconsistent datasets^{8 19 67 68} while two (2%) studies used the average of its nearest values.^{69 70} Other approaches used included the multivariate stepwise regression,⁷¹ using data values extracted from previous surveys.³²

Modelling covariates

The type and number of covariates included in malaria models varied across studies. Different categories encompassing climatic and environmental, sociodemographic and malaria intervention covariates were identified. The most common covariates in the environmental domain were rainfall and temperature used in 61 (57%) and 59 (55%) studies, respectively; while the most common sociodemographic covariates used in 12 (11%) studies were population size and age. Malaria interventions (insecticide-treated bed nets, indoor residual spraying and artemisinin-based combined therapy) used in 32 (30%) studies and transmission seasonality used in 28 (26%) studies were also common. Detailed variations and adaptations covariates are presented in table 3.

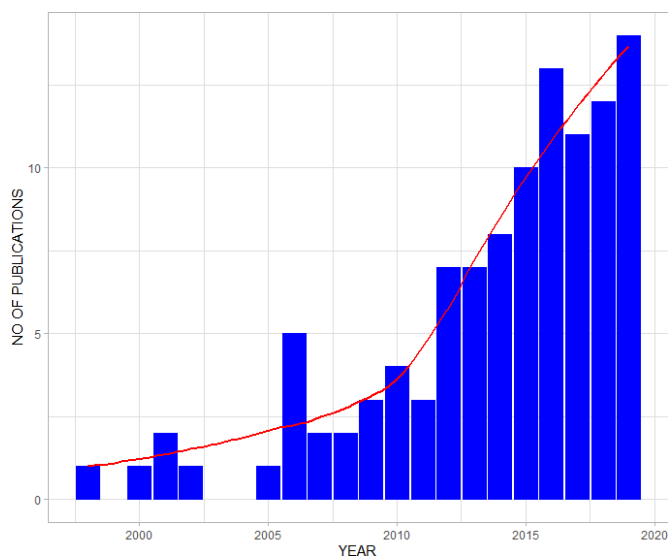


Figure 3 Bar—chart with a trend line (red) showing the total number of included studies.

Table 1 Data sources

Type	Source	No	References
Global/continental databases	Malaria Transmission Intensity and Mortality Burden across Africa	1	88
	Mapping Malaria Risk in Africa databases	9	29 31 44 45 63 67 73 97 98
	World Pop/Afripop	14	25 41 46–48 59 60 72 75 88 99–102
	Food and Agriculture Organisation-Food Security and Nutrition Analysis Unit	1	59
	Global Rural and Urban Mapping project	2	41 99
	WHO database on malaria drug resistance	1	98
	Global Lakes and Wetlands Database	3	34 48 49
	UN World Urbanisation prospects database	4	25 49 50 64
National databases	Health and Demographic Surveillance System	16	28 32 38 42 79 100 102–111
	Census	6	21 22 68 108 112 113
	National statistical agencies	10	40 51 65 69 107 114–118
	Demographic Health Survey	7	23 26 52 61 66 98 101
	Malaria Indicator Survey	12	37 41 43 49 52 60 66 75 76 99 117 119
Subnational databases	Cross-sectional surveys	9	8 20 46 48 49 53 120–122
	Cohort studies	5	33 39 54 84 123–125
	Cluster surveys	1	34
	Entomological/parasitological surveys	5	77 113 120 126 127
Remote sensing	Moderate Resolution Imaging Spectroradiometer	28	25 27 29 30 34 37 38 41 48 55 59–61 65 66 72 75–79 99 100 103 104 128–130
	Africa Data Dissemination Service	8	29 30 65 71 76–79
	United States Geological Survey-Earth Resources Observation and Science Centre	8	27 28 30 35 43 76 77 99
	Health Mapper	8	27 29 30 41 61 65 68 76 77
	Shuttle Radar Topographic Mission	5	28 60 72 99 129
	WorldClim-Global Climate database	7	23 34 37 59–61 100
	Tropical Rainfall Measuring Mission	3	104 128 130
	Early Warning System	3	66 72 88
	Climate Research Unit	3	23 71 131
	National Oceanic and Atmospheric Administration	2	109 132
	Water Resources Institute	1	63
	World Wildlife Fund	1	37
	Africover	1	34
	Famine Early Warning Systems Network Land Data Assimilation System	1	88
Ground station data	Meteorological data	17	21 22 35 38 40 54 69–71 84 103 106 113 126 133–135

Table 2 Analytical methods used in malaria risk mapping

Category	Method	No	References
Variable selection techniques	Stepwise procedures	11	20 28 34–36 44 45 61 67 73 135
	Preliminary frequentist analysis	14	21 22 24–27 29–32 61 63 84 134
	Total-set analysis	6	8 48 49 58–60
	Principal component analysis	6	32 39–43
	Bayesian stochastic search	3	72 77 119
	LASSO penalty	2	47 88
	Literature review	2	30 74
	Spike and slab	2	72 99
	BMA	1	100
Visualisation	Rate map	63	8 19 24 26–32 34 35 37 40 41 43 45 46 49–51 53 55 57–68 71–80 84 88 97 98 100–102 108 109 111 112 114 117 119 122 124 135 136
	Dot map	25	20 27 29 31 36 39 42 48 50 55–57 63 65 72 99 105 113 120 121 123 125 127 132 137
	Case counts	16	21 25 41 44 54 65 66 72 76–78 98 106 115 118 131
Spatial cluster ‘hotspot’ analysis	Spatial scan statistic	15	33 38–40 42 51 54 105 114 120 122 123 127 134 137
	Global Moran’s/	6	23 56 57 68 115 121
	Getis Ord statistic	3	32 112 113
	Local Moran’s/	7	23 32 68 102 108 117 118
Spatial/spatio-temporal modelling	Geostatistical models	27	8 26 27 29–31 34 36 37 41 43 46 48–50 53 59–61 63–65 74–78
	Bayesian CAR models	15	19 21 22 24 25 44 47 66 84 102 103 119 129 131 135
	Time series models	9	40 51 69 70 107 125 128 130 133
	Bayesian Kriging	5	45 62 67 72 73
	Conventional Poisson	7	97 104 109–112 126
	Conventional logistic	4	28 35 52 113
	GAM	2	38 40
	Negative binomial regression	1	117
	GWR	1	57
	ANN	1	116
	BRT	1	55
Model validation/predictive ability	Data partitioning	24	8 19 27 34 36 37 41 46 48–50 53 58 61 63–65 76–78 80 88 109 110
	Deviance information criterion	19	21–25 30 34 60 64 66 75 79 84 102 103 110 125 131 135
	Akaike information criterion	8	37 60 66 75 79 113 128 130
	Root mean squared error	7	35 47 55 64 79 88 130
	Variogram-based algorithm	7	19 47 72 79 84 99 119
	Mean absolute prediction error	6	8 48 49 58 59 61
	Mean error	3	34 36 47
	Bayesian information criterion	2	23 107

ANN, Artificial neural network; BMA, Bayesian model averaging; BRT, Boosted regression tree; CAR, Conditional autoregressive; GAM, General additive model; GWR, Geographically weighted regression; LASSO, Least absolute shrinkage and selection operator.

Spatial, temporal and spatio-temporal methods

A variety of spatial, temporal and spatio-temporal methods were employed to visualise malaria risk patterns, explore spatial clusters and model risk across space and time in SSA. Measurement of malaria burden varied across studies with the type of outcome informing the modelling framework. The most common malaria metric used in models, was incidence used in 50 (47%) studies and

prevalence used in 37 (35%) studies. Table 3 presents a summary of the malaria outcomes that were considered in the papers included in the review.

In settings of low malaria transmission, local and global spatial cluster detection methods were used in 31 (29%) studies to identify significant geographical variation in malaria risk patterns (table 2). These were the Kulldorf spatial scan statistic, Getis' $G_i^*(d)$ local statistic;

Table 3 Covariates used in malaria risk mapping

Indicator	Metric	No	References
Malaria Outcome	Malaria incidence/cases	50	21–25 33 38–40 47 51 54 57 62 64 66 68 70–73 84 88 97 102 105 107–110 113–119 123 125 127–131 133–138
	Malaria prevalence	37	8 19 20 26 29 31 34 36 37 42 44–46 48–50 52 53 55 56 58–61 67 73–75 99 100 112 120–122 124 126 132
	Malaria risk	12	27 28 30 35 41 43 63 65 76–78 101
	Malaria mortality/deaths	5	32 69 98 103 104
	EIR/Estimate/Mosquito density/abundance	3	79 80 106
Rainfall indices	Rainfall/precipitation	44	21 22 27–31 34 35 40 44 45 47 55 57 58 60 63 64 66 67 69–72 75–78 80 100 101 103 104 110 113 116 118 119 128 129 132 134 135
	Monthly rainfall	10	25 73 97 102 107 109 126 130 131 133
	Annual rainfall	5	23 24 48 49 59
	Weekly rainfall	2	54 88
Temperature indices	TSI	10	8 25 37 48 49 58–60 80 110
	LST	19	27 29 30 39 47 65 72 76–79 99–101 104 110 119 129 130
	Mean/min/max temperature	28	21 22 24 31 36 44 45 55 57 63 66 67 69 70 73 97 102 103 106 109 116 118 128 131–135
	Weekly temperature	2	40 88
Vegetation indices	NDVI	31	24 27 29 30 35 38 43–45 47 55 63 65 67 71–73 77 79 80 100 101 103 104 109 119 129–132 135
	EVI	17	8 29 34 36 39 47–49 58–60 64 66 75 110 128 130
	Annual EVI	2	25 64
	Monthly EVI	1	37
	Leaf area index	1	29
GIS Derived	Distance to nearest water source	34	20 23 27 29 30 33–37 44 47–49 55 58 59 61 63 65 67 75 76 79 80 97 100 106 113 115 117 119 129 134
	Distance to main road	6	28 35 61 100 115 117
	Distance to health facility	4	32 35 61 100
	Distance to urban centre	2	57 61
	Distance to border	1	97
Elevation	Altitude	10	20 36 55 72 75–77 101 119 129
	Elevation	11	28 30 31 35 37 43 60 78 79 100 135
Land cover	Land cover	8	47 55 67 72 77 78 99 119
Humidity	Relative humidity	8	21 22 40 60 69 70 106 118
	Weekly humidity	1	88
	Evapotranspiration	2	31 130
	Vapour pressure	3	24 88 131
	Evaporation	1	69
Digital Elevation Models - DEM derivatives	Wetness index/CTI	2	37 55
	Slope	5	28 35 37 61 100
	TWI	1	47
	Aridity index	1	37
Reflectivity	Stable lights	1	37
	Visibility	1	70
Wind	Wind speed	3	40 70 118

Continued

Table 3 Continued

Indicator	Metric	No	References
Demographic factors	SES	9	26 32 42 43 99 100 102 108 122
	Gender/Sex	6	26 32 57 84 100 113
	Age	12	26 32 42 43 57 75 84 99 100 103 106 113
	Population density/size	12	25 37 45 57 67 75 98 100 112 115 117 135
	Livestock ownership	2	42 57
	Urbanisation	8	8 26 36 48 58–60 72
	Development	1	113
	Wealth index/category	4	75 99 119 129
	Building/Housing material	4	42 57 100 123
Time	Year/Month of survey	3	34 115 126
	Time period	1	84
	Transmission seasonality	28	25 28 32 35 37 40 44 51 59 60 63 67 69 70 80 84 100 101 104 113 115–117 119 125 129–131
Malaria intervention	ITN/LLIN ownership/coverage/use	19	32 33 42 43 57 66 74 75 99 100 103 112 113 119 122 123 126 129 134
	IRS	8	26 33 74 75 99 100 112 117
	ACTs	5	74 98 99 119 129
	Treatment seeking rate	3	66 119 129
	Reporting and testing	1	66
None	None	19	19 46 50–53 56 62 68 105 114 120 121 124 125 127 136–138

ACTs, Artemisinin-based combined therapy; CTI, Compound topographic index; DEM, Digital elevation models; EIR, Entomological inoculation rate; EVI, Enhanced vegetation index; GIS, geographical information system; IRS, Indoor residual spraying; ITN, Insecticide-treated bed nets; LLIN, Long lasting insecticidal nets; LST, Land surface temperature; NDVI, Normalised difference vegetation index; NDWI, Normalised difference water index; SES, Social economic status; TSI, Temperature suitability index; TWI, Topographic wetness index.

local Moran's I statistic and the Global Moran's I statistic. On the other hand, nine (8%) studies used temporal models to explore and forecast malaria risk at different temporal resolutions, with the autoregressive integrated moving average (ARIMA) model used in seven (6%) studies being the most common. Two studies (2%) used a univariate seasonal ARIMA model to explore malaria risk patterns (table 2).

The Bayesian spatial only and space-time kriging—a statistically unbiased and robust interpolation method appropriate for study settings with limited data; was used in five (6%) studies, to predict risk at unsampled locations,^{67 72} improve predictions in geographical areas with considerable variation between observed values and model predictions^{45 73} and model the spatial and temporal correlations of monthly malaria morbidity cases.⁶²

Using point-referenced data sourced from multiple independent surveys, 27 (25%) studies applied both the model-based geostatistical (MBG) and Bayesian MBG methods to analyse, predict and map malaria risk. In this framework, the spatio-temporal dependency was modelled as a Gaussian process in fourteen (13%) studies.^{8 34 36 37 43 46 48–50 53 59 60 64 74} Spatial only random effects (dependency) were modelled via the Gaussian prior distribution^{26 27 29–31 41 63 65 75–78} in 12 (11%) studies. Temporal random effects were assigned a first order

autoregressive AR (1) prior distribution in three (3%) studies^{37 75 79} and second order autoregressive AR(2) prior distribution⁸ in one (1%) study (table 4).

Using observations aggregated over distinct geographical region/spatial partitions/adjacent units (eg, census tract, administrative boundaries); 15 (14%) studies used the Bayesian conditional autoregressive (CAR) models, to explore the spatial and spatio-temporal variation of malaria risk. To account for the temporal dependency between consecutive time points; seven studies (6%) used the first order autoregressive AR (1) prior process, whereas one study (1%) used the random walk of order one RW (1) prior process (table 4).

Other models

Generalised linear modelling framework, such as the Poisson, logistic regression, negative binomial and geographically weighted regression, was used in fifteen studies (14%). These models explored the association of malaria counts or rates and its correlates, using appropriate exponential distribution families. Machine learning techniques such as the artificial neural network and the boosted regression tree were used to analyse incidence patterns and to examine malaria prevalence, respectively (table 4).

Model validation, performance and uncertainty

A range of different validation techniques were used to assess model fitness and to select the optimal predictive

Table 4 Structure of the spatio-temporal models

ID	References	Year	Space	Time	Space time
1	Abellana <i>et al</i> ⁸⁴	2008	CAR		
2	Alegana <i>et al</i> ⁶⁴	2016	Markov random field	–	Gaussian
3	Alegana <i>et al</i> ²⁵	2013	–	–	CAR
4	Alemu <i>et al</i> ⁵¹	2013	–	Temporal trend – ARIMA	–
5	Amek <i>et al</i> ⁷⁹	2012	Gaussian	AR (1)	–
6	Amratia <i>et al</i> ⁶¹	2019	Gaussian	–	–
7	Appiah <i>et al</i> ⁶²	2011	–	–	STOK
8	Awine <i>et al</i> ¹³³	2018	–	–	SARIMA
9	Bejon <i>et al</i> ³⁹	2010	Cluster analysis	Temporal trends	–
10	Bejon <i>et al</i> ¹⁰⁵	2014	Cluster analysis	–	–
11	Belay <i>et al</i> ¹⁰⁶	2017	–	Temporal trends	–
12	Bennett <i>et al</i> ⁶⁰	2013	–	–	Gaussian
13	Bennett <i>et al</i> ⁷⁵	2016	Gaussian	–	–
14	Bennett <i>et al</i> ⁶⁶	2014	CAR	CAR	CAR
15	Bhatt <i>et al</i> ⁷⁴	2015	Markov random field	AR (1)	Gaussian
16	Bisanzio <i>et al</i> ¹¹³	2015	Markov random field	B – splines with RW (2)	–
17	BM & OE ⁴⁴	2007	CAR	–	–
18	Bousema <i>et al</i> ¹²³	2010	Hotspot analysis	–	–
19	Ceccato <i>et al</i> ⁷¹	2007	Cluster analysis	–	–
20	Chipeta <i>et al</i> ⁵⁰	2019	–	–	Gaussian
21	Chirombo <i>et al</i> ¹⁰⁹	2020	Markov random field	Markov random field	Gaussian
22	Cissoko <i>et al</i> ³⁸	2020	Cluster analysis	Temporal trend	–
23	Colborn <i>et al</i> ⁸⁸	2018	–	–	Gaussian
24	Coulibaly <i>et al</i> ⁵⁴	2013	Cluster analysis	–	–
25	DePina <i>et al</i> ¹¹⁸	2019	Cluster analysis	Temporal trend	–
26	Diboulo <i>et al</i> ⁴¹	2016	Gaussian	–	–
27	Ferrão <i>et al</i> ⁷⁰	2017a	–	Temporal trend - ARIMA	–
28	Ferrão <i>et al</i> ⁶⁹	2017b	–	Temporal trend - ARIMA	–
29	Ferrari <i>et al</i> ¹²⁴	2016	Cluster analysis	–	–
30	Gaudart <i>et al</i> ¹²⁵	2006	Cluster analysis	Temporal trend - ARIMA	–
31	Gemperli <i>et al</i> ⁶⁷	2006	Exponential correlation function	–	–
32	Gething <i>et al</i> ⁹⁸	2016	–	P – splines with RW (1)	–
33	Giardina <i>et al</i> ⁷⁸	2015	Gaussian	–	–
34	Giardina <i>et al</i> ⁶⁵	2012	Multivariate Normal	–	–
35	Giardina <i>et al</i> ¹⁰¹	2014	Gaussian	–	–
36	Giorgi <i>et al</i> ⁴⁶	2018	–	–	Gaussian
37	Gómez-Barroso <i>et al</i> ²⁰	2017	Cluster analysis	–	–
38	Gosoni <i>et al</i> ²⁷	2012	Gaussian	–	–
39	Gosoni <i>et al</i> ⁷⁶	2010	Gaussian	–	–
40	Gosoni <i>et al</i> ⁶³	2006	Gaussian	–	–
41	Houngbedji <i>et al</i> ⁷⁷	2016	Normal	–	–
42	Ihantamalala <i>et al</i> ¹¹⁴	2018	Cluster analysis	–	–
43	Ikeda <i>et al</i> ¹¹⁶	2017	–	–	SOM
44	Ishengoma <i>et al</i> ¹²⁶	2018	–	Temporal trends	–

Continued

Table 4 Continued

ID	References	Year	Space	Time	Space time
45	Kabaghe <i>et al</i> ⁴³	2017	Gaussian	–	–
46	Kabaria <i>et al</i> ⁵⁵	2016	–	–	BRT
47	Kamuliwo <i>et al</i> ¹¹⁷	2015	Cluster analysis	–	–
48	Kang <i>et al</i> ³⁷	2018	Gaussian	AR (1)	–
49	Kangoye <i>et al</i> ¹²⁷	2016	Cluster analysis	–	–
50	Kanyangarara <i>et al</i> ²⁸	2016	–	–	–
51	Kazembe <i>et al</i> ³¹	2006	Gaussian	–	–
52	Kifle <i>et al</i> ¹⁰⁷	2019	Cluster analysis	Temporal trends - SARIMA	–
53	Kigozi <i>et al</i> ¹²⁸	2016	–	Temporal trend- ARIMA	–
54	Kleinschmidt <i>et al</i> ⁴⁵	2000	Kriging	–	–
55	Kleinschmidt <i>et al</i> ⁷³	2001a	Kriging	–	–
56	Kleinschmidt <i>et al</i> ⁹⁷	2001b	Kriging	–	–
57	Kleinschmidt <i>et al</i> ¹³⁶	2002	Normal	Normal	–
58	Mabaso <i>et al</i> ¹³¹	2005	CAR	–	AR (1)
59	Mabaso <i>et al</i> ²⁴	2006	CAR	AR (1)	–
60	Macharia <i>et al</i> ⁵³	2018	–	–	Gaussian
61	Mfueni <i>et al</i> ⁵²	2018	–	–	–
62	Midekisa <i>et al</i> ¹³⁰	2012	–	Temporal trend - SARIMA	–
63	Millar <i>et al</i> ¹⁰⁰	2018	–	–	–
64	Mirghani <i>et al</i> ¹²⁰	2010	Cluster analysis	–	–
65	Mlacha <i>et al</i> ⁵⁶	2017	Cluster analysis	–	–
66	Mukonka <i>et al</i> ¹³⁸	2014	–	Temporal trends	–
67	Mukonka <i>et al</i> ¹¹⁵	2015	Cluster analysis	–	–
68	Mwakalinga <i>et al</i> ¹²¹	2016	Cluster analysis	–	–
69	Ndiath <i>et al</i> ⁵⁷	2015	Cluster analysis	–	–
70	Ndiath <i>et al</i> ⁴²	2014	Cluster analysis	–	–
71	Nguyen <i>et al</i> ¹¹⁰	2020	Gaussian	–	Gaussian
72	Noor <i>et al</i> ⁴⁸	2013a	Gaussian	–	GRF
73	Noor <i>et al</i> ³⁴	2008	Gaussian	–	–
74	Noor <i>et al</i> ⁵⁸	2012b	–	–	GRF
75	Noor <i>et al</i> ³⁶	2009	–	–	GRF
76	Noor <i>et al</i> ⁸	2014	Gaussian	AR (2)	–
77	Noor <i>et al</i> ⁴⁹	2013b	–	–	GRF
78	Noor <i>et al</i> ⁵⁹	2012a	Gaussian	–	Stationary Gaussian
79	Nyadanu <i>et al</i> ¹⁰⁸	2019	Cluster analysis	–	–
80	Okunola <i>et al</i> ²³	2019	Cluster analysis	–	–
81	Onyiri ²⁹	2015	Gamma	–	–
82	Ouedraogo <i>et al</i> ⁴⁰	2018	–	Temporal trend- ARIMA	–
83	Ouédraogo <i>et al</i> ¹¹¹	2020	CAR	AR (1) / Temporal trends	–
84	Peterson <i>et al</i> ¹³⁴	2009	Cluster analysis	–	–
85	Pinchoff <i>et al</i> ³⁵	2015	–	–	–
86	Raso <i>et al</i> ³⁰	2012	Multivariate Normal	–	–
87	Rouamba <i>et al</i> ¹⁰²	2020	CAR	CAR	Gaussian
88	Rumisha <i>et al</i> ⁸⁰	2014	Gaussian	AR (1)	–

Continued

Table 4 Continued

ID	References	Year	Space	Time	Space time
89	Selemani <i>et al</i> ³²	2015	Cluster analysis	–	–
90	Selemani <i>et al</i> ¹⁰³	2016	CAR	AR (1)	–
91	Sewe <i>et al</i> ¹⁰⁴	2016	–	Natural cubic spline	–
92	Seyoum <i>et al</i> ¹³⁷	2017	Cluster analysis	–	–
93	Shaffer <i>et al</i> ¹²²	2020	Cluster analysis	Temporal trends	–
94	Simon <i>et al</i> ¹¹²	2013	Cluster analysis	–	–
95	Siraj <i>et al</i> ¹³⁵	2015	CAR	–	–
96	Snow <i>et al</i> ¹⁹	2017	CAR	CAR	–
97	Snow <i>et al</i> ¹³²	1998	–	–	–
98	Solomon <i>et al</i> ³³	2019	Cluster analysis	–	–
99	Ssempiira <i>et al</i> ¹¹⁹	2018a	CAR	AR (1) / temporal trend	–
100	Ssempiira <i>et al</i> ¹²⁹	2018b	CAR	AR (1) / temporal trend	–
101	Ssempiira <i>et al</i> ⁹⁹	2017b	CAR	–	–
102	Ssempiira <i>et al</i> ⁷²	2017a	–	–	–
103	Sturrock <i>et al</i> ⁴⁷	2014	CAR	Temporal trend	–
104	Yankson <i>et al</i> ²⁶	2019	Gaussian	–	–
105	Yeshiwondim <i>et al</i> ⁶⁸	2009	–	–	–
106	Zacarias and Andersson ²²	2011	CAR	AR (1)	–
107	Zacarias and Majlender ²¹	2011	CAR	RW (1)	–

AR, autoregressive; ARIMA, autoregressive integrated moving average; BRT, boosted regression tree; CAR, conditional autoregressive; GRF, Gaussian random field; RW, random walk; SARIMA, seasonal autoregressive integrated moving average; SOM, self-organising maps; STOK, space-time ordinary kriging.

models. The most commonly used approach entailed partitioning the data for model training and validation and was employed in 24 (22%) studies. The training set was then used to validate the predictive model fit, whereas the validation set was used for assessing the model predictive ability. The representative holdout datasets were selected using a spatially and temporal declustered algorithm,^{8 48 49 59} stratified sampling approach⁷⁵ and randomly.^{35 46 50 53 80} Information criteria, that is, the deviance information criterion, Akaike information criterion and the BIC were used in seventeen (16%) studies. Seven (7%) studies used variogram-based algorithms to identify estimates falling within the 95% credible interval. Model precision and accuracy metrics included the mean prediction error, root mean squared error, mean absolute prediction error, mean error and the SD (table 2).

Summarised modelling framework

The rapid expansion of methods and data informs the need to guide future spatial and spatio-temporal modelling of infectious diseases in SSA (online supplemental table 4). We illustrate a framework composed of four fundamental modelling entities, namely: inputs, process, stochastic components and output. Malaria data and covariates sourced from different spatial and temporal resolutions are considered to be the model inputs. A series of progressive and interdependent steps/processes on the model inputs are then used to generate the outputs

(posterior marginals). Posterior marginals can then be approximated using iterative computational techniques such as the Markov Chain Monte Carlo methods or by using numerical integrations via the Integrated Nested Laplace Approximations method.^{81 82} (figure 4)

DISCUSSION

Scalable guidelines for rigorous and transparent statistical methodology are necessary for reproducible malaria risk estimation. This review offers a comprehensive appraisal and synthesis of methods and covariates used in malaria risk mapping in SSA in the last five decades.

Sources of malaria data

High-resolution maps revealing the spatio-temporal variation of malaria endemicity are useful for estimating malaria burden, quantifying the effectiveness of control initiatives and assessing the progress towards its elimination nationally and subnationally. However, malaria risk mapping efforts in SSA are rarely based on routinely collected data. Instead, periodic and costly household survey's data have traditionally been used in modelling malaria risk. To address this challenge and obtain robust estimates reflective of the subnational burden, WHO initiated the high burden to high impact approach in 2018, which underscored the need for reliable national data systems. This is considered central to the understanding

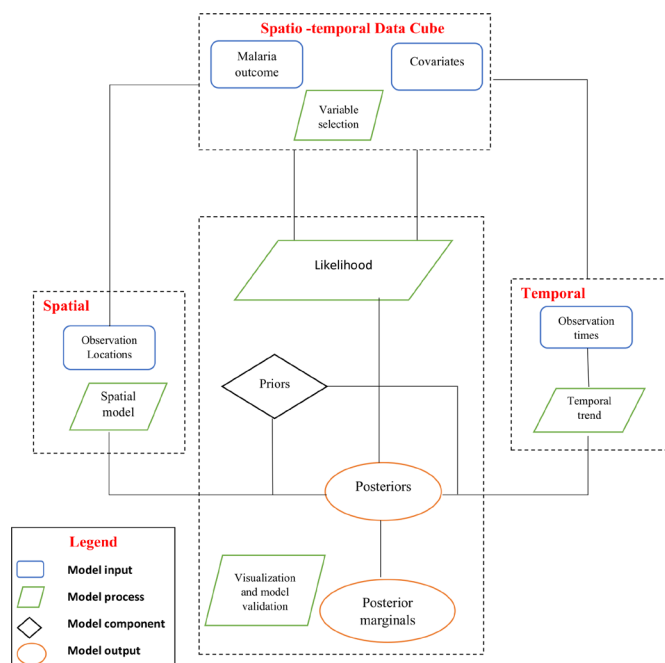


Figure 4 Schematic illustration of the spatio-temporal modelling framework for malaria risk in sub-Saharan Africa.

of malaria burden in low transmission settings and in the most vulnerable populations.³ Additionally, the approach has availed more malaria data in malaria-endemic settings, and caution is needed when gathering and interpreting findings generated from data at fine spatial and/or temporal scales with varying degree of completeness and representativeness.⁸³

The steady growth of satellite, remote sensing platforms and curated databases has made available a rich suite of both environmental and socioeconomic covariates at a finer level of detail useful for mapping malaria risk at high spatial and temporal resolution. Validating the quality of available satellite data prior to their inclusion in malaria studies remains central to achieving robust estimates.

While malaria incidence and prevalence metrics can be modelled from routine health information systems and sample surveys respectively, caution should be taken when interpreting estimates as both metrics are products of interacting factors such as interventions, sociodemographic and environmental factors that may contribute to the overall risk. A concise picture may be achieved by measuring malaria indicators at a finer spatial scale and exploring the nature and scope of the interaction. Data on malaria mortality as an outcome were sparse, and efforts must be made to increase data collection and improve the sensitivity and specificity of malaria mortality burden attribution in SSA.^{74 83} As many countries in SSA transition epidemiologically from high to low malaria transmission zones, obtaining useful metrics for mapping risk from sparse national surveys at low, moderate and heterogeneous transmission settings possess unique challenges for measuring progress and impact.⁸⁴

The paucity of continuous, reliable data necessary to yield estimates with greater geographical and temporal richness is a growing concern in the era of evidence-based public health. A high-quality, routinely collected data avail an alternative source of malaria metrics for continuous analysis over time.⁸³ Investments should be channelled towards establishing and addressing inadequacies in the health information systems to enable subnational mapping of malaria risk. However, the urge for quality data and the increasing need of accurate estimates can significantly be improved by adoption of data-driven modelling approaches that leverage both routine and household survey data in their model framework.⁸⁵ Additionally, detailed information on critical data sources, preliminary data adjustments undertaken before modelling should be availed as an important step towards enhancing reproducibility of methods and estimates.⁸⁶

Understanding covariates used in mapping malaria risk

Improving the precision of malaria risk estimates largely depends on limiting subjective decisions. These decisions may impact on the modelling process, even as more covariates becomes accessible at finer geographical and temporal resolutions. Studies have shown large variables to be desirable for prediction, whereas small sets of variables to be meaningful for inference.⁸⁷ It is important to understand the complex relationship between covariates at the same spatial or temporal resolution, to avoid overfitting. The trade-off between model interpretability, predictive ability, spatial and temporal scope, data accessibility and computational limitations are critical factors worth considering when selecting candidate covariates. Evidentially variable selection is an important preliminary step for a robust malaria risk mapping exercise. However, this approach continues to receive little attention amidst the growing diversity of covariate layers that need to be identified and included in the models.

Environmental and climatic factors influence mosquito vector abundance, distribution and longevity; at different time scales⁸⁸ and are important for mapping malaria risk.⁵⁹ A scoping review by Zinszer *et al*⁸⁹ highlighted the importance of climatic-related predictors in malaria risk prediction. Understanding the different facets and extent of how climatic influences on malaria risk variation; has been enhanced by advances in remote sensing and satellite imagery technology, increasing the availability of remotely sensed climatic data at high spatial resolution.³⁵ In this review proxies of; temperature (land surface temperature, temperature suitability index, mean/min/max/weekly) and rainfall/precipitation (weekly/monthly/annual) were the most widely used environmental factors related to malaria transmission in SSA. Vegetative indices such as elevation, surface moisture, land use and land cover were included in malaria risk maps, primarily due to their association with temperature and precipitation which indirectly influences⁹⁰ the distribution of malaria. Remote sensing will continue to feature prominently as a cost-effective tool

for mapping malaria risk in SSA and an important source of environmental and climatic covariates.⁹¹ The review further demonstrates the significance of non-climatic determinants such as malaria interventions and demographic factors in malaria risk mapping.

Modelling frameworks in malaria risk mapping

Complex decisions involving key modelling components such as covariates to include, preliminary data preprocessing and diagnostics checks demands advanced statistical knowledge. Extensive computational algorithms and complex spatio-temporal data structures may limit the applicability of these modelling approaches to experts. Furthermore, complex models used to represent malaria heterogeneity may not necessarily represent the truth on the ground. Thus, the statistical uncertainties around model estimates should be carefully examined, and the varying quantities and quality malaria data, that informs modelling approaches accounted for.

The review highlights the prominence and flexibility of geostatistical methods in modelling spatial and spatio-temporal malaria patterns, at policy-relevant units and thresholds.⁴⁶ Geostatistical methods provide a useful framework for interpolating imperfect data from multiple independent surveys by estimating spatial dependence from the data. At low spatial resolution, the Bayesian geostatistical framework accounts for uncertainty resulting from sparsely sampled point-referenced data by assigning priors that allows 'borrowing of strength' from adjacent regions leading to robust estimates and predictions.⁹² Amidst the current scarcity and imperfections of routine, high resolution and spatially expansive malaria data in many SSA countries, using geostatistical methods with data from multiple independent georeferenced surveys, continues to be important for generating reliable estimates.

Bayesian hierarchical CAR models are useful for modelling spatially correlated areal data by smoothing noisy estimates and leveraging information from adjacent regions. However, choosing an appropriate prior specification for the parameters defining the spatial interaction is inevitable and sometimes challenging. Notably, the spatial dependence among neighbouring regions is accounted for by assuming a CAR process in the random effects. For example, in the Besag York and Mollie/convolution model, location-specific spatial effects are assumed to follow a normal distribution with the mean equal to the average of its neighbours and the variance considered to be inversely proportional to the number of neighbours. In the Leroux *et al* model, the spatial dependence is based on the weighted average of both the independent random effects and spatially structured random effects.^{93 94} The intrinsic CAR and Besag, York and Mollie (BYM) were the most frequent global spatial smoothing specifications used in the review; given their easy implementation in a range of softwares. However, caution should be taken to minimise over smoothing—obscuring the underlying geographical patterns. Future modelling

studies should compare the impact of using other spatial smoothing priors.

Overall, malaria risk mapping has increased dramatically over the last decades, with novel methods advanced to meet the quest for accurate estimates of malaria burden. Whereas most approaches are built on classical statistical methods, recent advances in computing, availability of geographically referenced data have ushered/propagated new techniques designed to address existing challenges. These approaches include ensemble modelling, neural networks, simulation-based methods and bootstrap models to better capture space-time interactions.

Recommendations for best practices

As malaria landscape diversifies in the next decade, investments in primary data collection at subnational scales, development and continuous application of robust modelling tools will continue to be important priorities in malaria control and elimination efforts. In the era of open data policy and reproducible research, our review reiterates the importance of periodically reviewing, validating and updating malaria maps to accommodate new data sources, improved data quality, enhanced computing power and novel methodological approaches. Variable selection procedures should be data driven and objectively developed to the maximise the predictive accuracy of malaria risk mapping. The spatio-temporal modelling framework should incorporate practical challenges facing control and elimination of malaria in SSA. These challenges are: human migration within and among endemic zones, mapping asymptotic infection reservoirs and accounting for differential immunity within a population.^{95 96}

Strengths and limitations

The review search strategy was exhaustive and transparent, in accordance with the current methodological guidelines and included studies have provided a fair depiction of malaria risk mapping efforts in SSA. The methodological approach of the included studies was diverse, making meta-analysis inappropriate. The review considered only studies published in English and relevant papers published in other languages might have been excluded.

CONCLUSIONS

Malaria risk mapping remains an important component for understanding the burden of malaria in SSA. The review has described modelling approaches and examined covariates used in mapping malaria risk in different epidemiological contexts. As malaria transmission continues to decline in SSA, the use of metrics that accurately describes changes in its transmission intensity across space and time will be important for the design and implementation of evidence-based control and elimination measures.

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